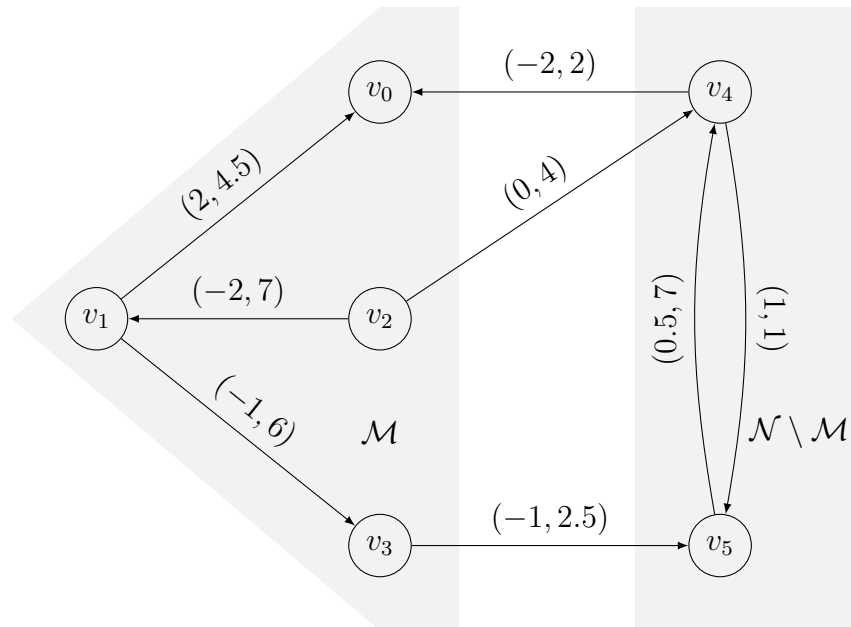


Networks: capacities

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Practice quiz

Consider the following network. Each arc (i, j) is associated with a pair of numbers (ℓ_{ij}, u_{ij}) , representing the lower and upper capacities, respectively. The gray areas are defining a cut.



1. Write the cut $\Gamma = \{\mathcal{M}, \mathcal{N} \setminus \mathcal{M}\}$ represented in the figure and the corresponding sets of forward arcs $\Gamma^{\rightarrow}$ and backward arcs $\Gamma^{\leftarrow}$.
2. Calculate the capacity of the cut Γ .
3. Assign a flow on each arc such that the cut is saturated.